course name : Computer Networks
course code : CSA0712
CO Assessed : CO1-Assessment Tool_1
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1. Email communication using OSI Model:

a) OSI layer.

1. physical layer:

Provides frame into Electrical, optical, or radio signals and transmits them through the physical medium.

2. Data link layers:

Converts packets into frames.

3. Network layers:

Adds Ip address and selects the best route for sending emails.

4. Transport layers:

It is heart of OSI layer and delivery across the TCP and UDP.

5. Session layers:

The sending emails are protected through or run by some protocols.

6. Presentation layer.

7. Application layer.

b) Data Encapsulation:

During transmission, each layer adds its own header.

Application → Message
Transport → segment.
Network → packet.
Datalink → Frame
physical - Bits

This process is called encapsulation.

c) De-encapsulation:

At the receiver:

- Physical receives bits.
- Data link removes frame header.
- Network removes IP header.
- Transport removes TCP header.
- Application displays the original email.

This process is called de-encapsulation.

2) Reliable File Transfer.

a) Transport Layer Reliability.

The Transport Layer (TCP):

- Divides the file into segments
- Uses sequence numbers
- Detects errors
- Retransmits last data.
- Ensures correct delivery.

b) Segmentation, Flow control, and Error control.

→ Splits a large file into smaller segments.

→ Prevents the sender from sending data too fast.

→ Detects and retransmits lost or corrupted segment.

c) Role of Acknowledgements (ACKs)

- The receiver sends an Ack after receiving each segment.
- If an Ack is not received, the sender retransmits the segment.
- This ensures reliable and error-free data transfer.

3. Data packet Transmission Across Multiple Networks.

a) Role of Network layer and Data Link layer

Network layer:

Adds Ip addresses and selects the best route for packet delivery using routers.

Data link layer:

Adds MAC addresses, creates frames, and transfers data between devices on the same network.

b) IP Addressing and MAC Addressing.

→ Identifies the source and destination across different networks.

→ Identifies devices within the local network and is used by switches to forward frames.

4. Errors During Data Transmission.

a) OSI Layers Responsible

- Datalink layer: Detects errors using CRC.
- Corrects errors by retransmitting lost or corrupted data using TCP.

b) Working Together.

- The Data link layer detects transmission errors.
- The transport layer requests retransmission of missing or damaged segments.
- Together they ensure reliable and accurate communication.

c) Example of Retransmission.

- A receiver detects a corrupted packet.
- It does not send an acknowledgement (Ack)
- The sender retransmits the packet.
- The receiver receives the correct packet and sends an Ack.

5. Web Browser Request Using HTTP.

a) Contribution of Each OSI layer.

- Application: sends the HTTP request.
- presentation: Formats and encrypts data.
- Session: Establishes and manages the session.
- Transport: Uses Tcp for reliable delivery.
- Network: Routes packets using IP.
- Data Link: Frames the data using MAC Addresses.
- Physical: Transmits datas as bits.

b) Flow of Data units:

- Application : Data
- Transport : segment.
- Network : packt
- Data link : Frame.
- physical : Bits.

c) Server Response Reaches the client.

- The server processes the HTTP request.
- It sends the response the OSI layers.
- Routers forward the packets using IP addresses.